Tianhao Fu

tianhao.fu@mail.utoronto.ca | atatc.github.io | linkedin.com/in/tianhao-fu-957188279 | github.com/ATATC

Education

University of Toronto — Toronto, ON
Bachelor of Applied Science in Engineering Science (PEY Co-op) cGPA 3.92/4.00 — *Sep 2025 – May 2030*

Experience

Machine Learning Intern — Sep 2026 – Present
Vector Institute (FLARE Lab) — *Toronto, ON, Canada*
- Supervisor: Dr. Jun Ma
- Working on the FLARE 2026 Task 3 summary paper
- Working on autoMIL (multi-instance learning)

Research Assistant — Apr 2026 – Sep 2026
Shanghai Jiao Tong University — *Shanghai, China*
- Supervisor: Dr. Bin Sheng
- Worked on "Who Does Medical AI Serve? Hidden Institutional Logics Behind Clinical AI"

Student Researcher — Apr 2026 – Sep 2026
University Health Network (Princess Margaret Cancer Center) — *Toronto, ON, Canada*
- Supervisor: Dr. Jun Ma
- Delivered a mini-course on multimodal large language models: https://minicourse-mllm.atatc.me/
- Applied C-RadioV4-style multi-teacher distillation on the 35TB SLC-PFM Pathology dataset from three encoders: H-optimus-1, UNI2-H, and Virchow2
- Utilized feature clustering with summary/patch token extraction to reduce the wall time from years to weeks on only 4 H100 GPUs
- Explored zero-shot feature upsampling (NAF) on pathology foundation models
- Managed the MICCAI FLARE 2026 challenge
- Fine-tuned and evaluated MedGemma 1 and 1.5 on various tasks across classification, multi-label classification, detection, instance detection, cell counting, regression and report generation, using Compute Canada (SLURM) clusters, as one of the baselines
- Made skills for agents to remotely access MFA-controlled SLURM clusters through shared ControlMaster SSH sessions

Research Team Lead — Sep 2025 – Jul 2026
University of Toronto Machine Intelligence Student Team — *Toronto, ON, Canada*
- **[UNSURE@MICCAI2026]** The SegWithU Project (SegWithU: Uncertainty as Perturbation Energy for Single-Forward-Pass Risk-Aware Medical Image Segmentation)
- The AIP project (Automated Iterative Pseudo-labelling)

Team Lead — Jun 2022 – Present
Project Neura — *Toronto, ON, Canada*
- The Varest Project (Just Respect the Semantics: Image-Token Compression via Pre-LLM Semantic Variable-Resolution Tokenization and In-LLM Relevance Pruning)
- **[UNSURE@MICCAI2026]** The SegWithU Project (SegWithU: Uncertainty as Perturbation Energy for Single-Forward-Pass Risk-Aware Medical Image Segmentation)
- Developed MIP Candy (A Candy for Medical Image Processing), a fast-prototyping engine for complete experiment pipelines for medical image processing
- Developed LEADS (Lightweight Embedded Assisted Driving System), an end-to-end instrumentation, control, and analysis system for electric race cars
- Built Erbium, an infrastructure cloud compute platform with Docker

Research Intern — Aug 2024 – Sep 2025
Vector Institute (Bo Wang Lab) — *Toronto, ON, Canada*
- Supervisor: Dr. Bo Wang
- **[Lancet Oncology]** Ranked 4th in the PANORAMA Challenge
- Proposed a multi-head UNet for joint classification (PDAC) and segmentation of 3D pancreatic lesions
- **[IEEE TMI]** Ranked 4th in the SegSTRONG-C Challenge
- Proposed domain-targeted data augmentation methods that improve nnUNet's robustness against non-adversarial destruction (e.g. smoke, blood, ...)

Projects

Varest | *Python, PyTorch, Transformers, PEFT, Triton* Feb 2026 – Apr 2026

- https://varest.github.io
- Proposed a semantic image-token compression framework for vision-language models, combining object- and text-guided merging with learned decoder-stage pruning
- Reduced layer-averaged image-token counts by 94.4% while retaining 86.1–92.3% of full-token performance, averaged across evaluation metrics
- Evaluated three VLM backbones on 13 benchmarks, outperforming 15 compression baselines in mean performance retention at every evaluated token budget
- Conducted ablations of object/text localization, token allocation, pruning depth, and random/oracle selection to quantify component contributions and accuracy–compute tradeoffs
- Designed prompt-swap, selector-refinement, and region-deletion experiments, revealing a narrow decoder-depth window where prompt-conditioned selection benefits grounding
- Evaluated semantic preservation through region overlap and native text retention, and analyzed task-specific visual dependence across decoder layers and token budgets

[UNSURE@MICCAI2026] SegWithU | *Python, PyTorch, MONAI, TorchMetrics* Feb 2026 – Apr 2026

- https://segwithu.projectneura.org
- Proposed a novel single-forward-pass uncertainty estimation method that models uncertainty as perturbation energy
- Conducted experiments that show SOTA performance on ACDC, BraTS2024, and LiTS datasets, against 8 baselines

CIV102 Bridge Project | *Python, NumPy, SciPy, Matplotlib* Nov 2025 – Dec 2025

- Top 1 in the cohort and 2nd in the class of 2025
- Developed a complex cross-section composition and solving system
- Developed a bridge solver
- Calculated the optimal cross-section dimensions by turning the design into a COP (Convex Optimization Problem)

The AIP Project | *Python, PyTorch, CUDA, Docker* Sep 2025 – Feb 2026

- Proposed an innovative way to make pseudo-labelling more efficient and effective

MIP Candy | *Python, PyTorch, CUDA, Triton, SimpleITK, Plotly* Aug 2025 – Present

- https://mipcandy.projectneura.org
- Developed a framework that brings ready-to-use training, inference, and evaluation pipelines together with aesthetics, so users can focus on their experiments, not boilerplate
- Developed a visualization system that helps the user see the complex data
- Accelerated training with a preloading mechanism
- Accelerated sliding window custom CUDA kernels

LEADS | *Python, FastAPI, OpenCV, Tkinter, PySerial, Next.js* Nov 2023 – Sep 2025

- https://leads.projectneura.org
- Developed an onboard instrumentation system that displays data like the wheel speed
- Developed a webpage dashboard that remotely monitors the vehicles' status from the Pit crew
- Developed a multi-camera streaming and recording system
- Developed support for saving and replaying vehicle data
- Developed an efficient real-time data link using pure TCP/IP
- Developed a data analysis tool set that teaches the driver how to drive
- Customized the Ubuntu OS
- Developed support for multiple screens
- Developed GPS support

Technical Skills

Languages: Python, C/C++, CUDA, JavaScript/TypeScript, HTML/CSS
ML/Scientific: PyTorch, Transformers, PEFT, MONAI, TorchMetrics, Triton, NumPy, SciPy, SimpleITK
Systems/Web: Docker, FastAPI, OpenCV, Tkinter, PySerial, React.js, Next.js, Plotly